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Inventor(s)

Michener; Jason Wallace et al.

SHOCK SERVICE ADVISORY ALGORITHM

Abstract

A method for determining an operation state of a suspension damper of a vehicle that may include receiving first data, determining whether the vehicle may be in a steady state condition based on the first data, and responsive to the vehicle being determined to be in the steady state condition, determining damping oscillation magnitude data for a set of suspension dampers based on second data. The method may further include performing a comparison between the damping oscillation magnitude data for a reference suspension damper and the damping oscillation data for remaining suspension dampers, determining an operation state of the reference suspension damper based on the comparison, cycling the reference suspension damper to one of the remaining suspension dampers, and repeating the method until the operation state of each one of a set of suspension dampers has been determined.

Inventors: Michener; Jason Wallace (Novi, MI), Dempsey; Rick (Milford, NH)

Applicant: Ford Global Technologies, LLC (Dearborn, MI)

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Background/Summary

TECHNICAL FIELD

[0001] Example embodiments generally relate to the operation of a vehicle's control system and, more particularly, relate to a control system for determining an operation state of suspension damper.

BACKGROUND

[0002] Modern vehicles have begun to utilize semi-active suspensions to continuously modify damping levels of a vehicle's suspension system to adjust to varying road conditions. These semi-active suspensions use suspension dampers to modify the damping level. However, detecting and classifying suspension damper faults is presently difficult. Due to the importance of damping adjustment, it would be useful to be able to detect and classify suspension damper faults quickly and early to ensure accurate suspension adjustment.

[0003] Thus, it may be desirable to be able to quickly and accurately define and classify suspension damper faults in order to notify a vehicle operator or service technician.

BRIEF SUMMARY OF SOME EXAMPLES

[0004] In accordance with an example embodiment, a control system for a suspension system of a vehicle may be provided. The control system may include a set of suspension dampers that may be operably coupled to a set of wheel assemblies of the vehicle, a first sensor suite that may measure first data that may include driver input data and vehicle system data, a second sensor suite that may measure second data that may include suspension assembly data, and a controller that may store the first data that may be received from the first sensor suite and the second data that may be received from the second sensor suite. The set of suspension dampers may include a reference suspension damper that may be operably coupled to a corresponding one of the set of wheel assemblies and remaining suspension dampers that may be operably coupled to remaining corresponding ones of the set of wheels. The controller may be configured to receive the first data, determine whether the vehicle may be in a steady state condition based on the first data, and responsive to the vehicle being determined to be in the steady state condition, determine damping oscillation magnitude data for the set of suspension dampers based on the second data. The controller may be further configured to perform a comparison between the damping oscillation magnitude data for the reference suspension damper and the damping oscillation data for the remaining suspension dampers, determine an operation state of the reference suspension damper based on the comparison, cycle the reference suspension damper to the one of the remaining suspension dampers; and repeat performing the comparison, determining the operation state of the reference suspension damper, and cycling of the reference suspension damper until the operation state of each one of the set of suspension dampers has been determined.

[0005] In another example embodiment, a method for determining an operation state of a suspension damper of a vehicle may be provided. The method may include receiving first data that may include driver input data and vehicle system data that may be measured from a first sensor suite, determining whether the vehicle may be in a steady state condition based on the first data, and responsive to the vehicle being determined to be in the steady state condition, determining damping oscillation magnitude data for a set of suspension dampers based on second data that may include suspension assembly data that may be measured from a second sensor suite. The method may further include performing a comparison between the damping oscillation magnitude data for a reference suspension damper and the damping oscillation data for remaining suspension dampers, determining an operation state of the reference suspension damper based on the comparison, cycling the reference suspension damper to one of the remaining suspension dampers, and repeating performing the comparison, determining the operation state of the reference suspension damper, and cycling of the reference suspension damper until the operation state of each one of a set of suspension dampers has been determined. The set of suspension dampers may be operably coupled to a set of wheel assemblies of the vehicle. The set of suspension dampers may include the

reference suspension damper that may be operably coupled to a corresponding one of the set of wheel assemblies and the remaining suspension dampers that may be operably coupled to remaining corresponding ones of the set of wheels.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0006] Having thus described the invention in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0007] FIG. 1 illustrates a block diagram of a vehicle control system in accordance with an example embodiment;

[0008] FIG. 2 illustrates a flow chart of an algorithm for detecting an operation state of a set of suspension dampers in accordance with an example embodiment;

[0009] FIG. 3 illustrates a flow chart for determining if a fault state may be occurring in accordance with an example embodiment;

[0010] FIG. 4 illustrates a flow chart for cycling a reference suspension damper in accordance with an example embodiment;

[0011] FIG. 5 illustrates a flow chart for classifying the fault state; and

[0012] FIG. 6 illustrates a block diagram of a method in accordance with an example embodiment.

DETAILED DESCRIPTION

[0013] Some example embodiments now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all example embodiments are shown. Indeed, the examples described and pictured herein should not be construed as being limiting as to the scope, applicability or configuration of the present disclosure. Rather, these example embodiments are provided so that this disclosure will satisfy applicable requirements. Like reference numerals refer to like elements throughout. Furthermore, as used herein, the term “or” is to be interpreted as a logical operator that results in true whenever one or more of its operands are true. As used herein, operable coupling should be understood to relate to direct or indirect connection that, in either case, enables functional interconnection of components that are operably coupled to each other.

[0014] FIG. 1 illustrates a block diagram of a control system **100** according to an example embodiment. The components of the control system **100** may be incorporated into a vehicle **110** (e.g., may be operably coupled to a chassis of the vehicle **110**, various components of the vehicle **110** and/or electronic control systems of the vehicle **110**). In some cases, the chassis may include or be defined by a frame, and the frame may additionally be formed of one or more casted subframes. Of note, although the components of FIG. 1 may be operably coupled to the vehicle **110** and to each other, it should be appreciated that such connection may be either direct or indirect. Moreover, some of the components of the control system **100** may be connected to the vehicle **110** via intermediate connections to other components either of the chassis or of other electronic and/or mechanical systems or components.

[0015] The control system **100** may include a set of wheel assemblies **120**. For example, in an example embodiment, the set of wheel assemblies **120** may include a front left wheel assembly, a rear left wheel assembly, a front right wheel assembly, and a rear right wheel assembly. The set of wheel assemblies **120** may be operably coupled to a set of suspension dampers **160**. In some cases, the set of wheel assemblies may include a reference suspension damper that may be operably coupled to a corresponding one of the set of wheel assemblies **120** and may include remaining suspension dampers that may be operably coupled to remaining corresponding ones of the set of wheel assemblies **120**. In some cases, the set of wheel assemblies **120** may include various additional suspension system components which may include corresponding shocks, motion sensors for measuring rate of change of velocity, wheel bearings, brakes assemblies, and wheel

hubs that are operably coupled to the corresponding ones of the set of wheel assemblies **120**. In an example embodiment, the set of suspension dampers **160** may include any number of single instances of a suspension damper. In some cases, the set of wheel assemblies **120** may include any number of single instances of a wheel assembly. In an example embodiment, the set of wheel assemblies **120** and the set of suspension dampers may be part of a suspension assembly of the vehicle **110**.

[0016] The vehicle **110** may include a first sensor suite **115**. In an example embodiment, the first sensor suite **115** may include accelerometers, inertia sensors, driver input sensors, vehicle monitoring sensors, or any number of sensors that provide information on the operational state of the vehicle. In some cases, the inertia sensor may be an inertial measurement unit (IMU) using a plurality of signals from a plurality of sensors (accelerometers, position sensors, etc.) to create a vehicle dynamic model (VDM) to encapsulate the vehicle state. In an example embodiment, the driver input sensors may include throttle input sensors, brake input detectors, steering input detectors, or any number of input detectors for detecting the change in vehicle operator inputs. In some cases, the vehicle monitoring sensor may include tire pressure sensors, component sensors, or any number sensors that ensure components of the vehicle **110** are within the desired operation conditions. In an example embodiment, the first sensor suite **115** may be used for measuring first data. In some cases, first data may include driver input data, vehicle system data, or any other data that may be outputted from sensors within the first sensor suite **115**.

[0017] In an example embodiment, the vehicle **110**, the set of wheel assemblies **120**, or the set of suspension dampers **160** may include a second sensor suite **165**. The second sensor suite **165** may measure second data, which may be indicative of suspension assembly data such as suspension height data, rate of change of suspension position (suspension velocity), or rate of change of suspension velocity. Suspension assembly data may be any data that is representative of a state of the suspension assembly or a state of any suspension assembly component. In some cases, the second sensor suite **165** may include suspension height (or position) sensors, accelerometers, or any number of sensors that may detect suspension assembly data. In an example embodiment, the suspension height sensor may measure suspension height data, which may include suspension height, vehicle or wheel assembly ride height, wheel assembly position, or various other parameters of set of wheel assemblies **120**. The various other parameters of the set of wheel assemblies **120** may be parameters such as velocity or rate of change of velocity that may be derived from position data, which may be measured by the suspension height sensor. The suspension height sensor may be a rotary suspension height sensor that may utilize a rotary arm to measure suspension height data. In some cases, the suspension height sensor may be a linear position sensor which may be operably coupled to a suspension damper. In an example embodiment, the linear position sensor may be a Hall sensor, which may utilize the Hall effect to determine suspension height data of the corresponding ones of the set of wheel assemblies **120**. The linear position sensor may be an inductive sensor, hall sensor, or sensor/any other sensing technology used to determine suspension movement (either position or rate of change of velocity). In some cases, the suspension height sensor may be any type of sensor that may measure suspension height data of the corresponding ones of the set of wheel assemblies **120**. In an example embodiment, the suspension height sensor may measure voltage measurements as the suspension height data and may correlate the voltage measurements to respective wheel positions along a vertical axis between full compression and full rebound of the wheel assemblies to determine the suspension height data.

[0018] Additionally, the control system **100** may further include a controller **200**. While the controller **200** may have many functions, one such function of the controller **200** may be monitoring the status of other components of the vehicle **110**. The components that the controller **200** may monitor the status of may be included in, or not included in, the control system **100**. In this regard, the controller **200** may be responsible for detecting any faults associated with the components of the vehicle **110**. In an example embodiment, the controller **200** may receive

information that may be used to determine the status of various components or subassemblies of the vehicle **110** via various sensors that may be operably coupled to the components or subassemblies. Such sensors may be part of a sensor network and sensors of the sensor network may be operably coupled to the controller **200** (and/or the components or subassemblies). In some cases, the controller **200** may communicate the status of certain components in the vehicle **110** to other components in the vehicle **110**, so that appropriate action may be taken to ensure the vehicle **110** may continue operating without any components that may be experiencing a fault.

[0019] In some cases, the controller **200** may include a comparison module **210** and a communication module **220**. The comparison module **210** may be primarily responsible for determining if a fault may be occurring within the control system **100**. The communication module **220** may be responsible for communicating information internally within the vehicle **110** or externally to other controllers or devices. The communication module **220** may communicate wirelessly, via a wired connection, or via any other communication protocol, method, or technique that may provide reliable communication between the controller **200** and the target device or destination. In an example embodiment, the communication module **220** may communicate with a user interface **221** of the vehicle **110** or with a user device of a service technician **222**. In some cases, the controller **200** may store the first data and the second data. In an example embodiment, the controller **200** may include processing circuitry.

[0020] The processing circuitry may be configured to provide electronic control inputs to one or more functional units of the control system **100** and to process data received at or generated by the one or more functional units of the control system **100**. Thus, the processing circuitry may be configured to perform data processing, control function execution and/or other processing and management services according to an example embodiment. In some embodiments, the processing circuitry may be embodied as a chip or chip set. In other words, the processing circuitry may comprise one or more physical packages (e.g., chips) including materials, components and/or wires on a structural assembly (e.g., a baseboard). The structural assembly may provide physical strength, conservation of size, and/or limitation of electrical interaction for component circuitry included thereon. The processing circuitry may therefore, in some cases, be configured to implement an embodiment of the present invention on a single chip or as a single “system on a chip.” As such, in some cases, a chip or chipset may constitute means for performing one or more operations for providing the functionalities described herein.

[0021] In an example embodiment, the processing circuitry may include one or more instances of a processor and memory that may be in communication with or otherwise control other components or modules that interface with the processing circuitry. As such, the processing circuitry may be embodied as a circuit chip (e.g., an integrated circuit chip) configured (e.g., with hardware, software or a combination of hardware and software) to perform operations described herein. In some embodiments, the processing circuitry may be embodied as a portion of an onboard computer housed in the housing to control operation of the control system **100**.

[0022] FIG. 2 illustrates a flow chart of an algorithm **300** for detecting an operation state of a set of suspension dampers **160**. FIG. 3-FIG. 5 illustrate more detailed flow charts and examples for operations within the algorithm **300** in accordance with an example embodiment. The algorithm **300** may be run by the controller **200**. The algorithm **300** initially may begin with configuration the first sensor suite **115**, the second sensor suite **165**, and any other sensors or systems within the control system **100**, as well as may perform inventory of components and systems within the control systems in operation **301**.

[0023] The algorithm **300** may determine vehicle parameters at operation **310**. In some cases, the vehicle parameters may be parameters measured by the first sensor suite **115** and represented by the first data. In an example embodiment, the vehicle parameters may be one or more vehicle parameters determined by operation **310**.

[0024] After the vehicle parameters are determined, the algorithm **300** may determine whether the

vehicle is in a steady-state condition based on the vehicle parameter at operation **315**. The steady-state condition may be defined as occurring when the vehicle parameters are detected as constant for a steady-state determination time period. In some cases, the vehicle **110** may not have a vehicle parameter be constant for an entire period of the steady-state determination time period, in which the steady-state condition is defined as occurring when the vehicle parameters are relatively constant (vehicle parameters values within a certain range or percentage of one another) for the steady-state determination time period. For instance, one of the vehicle parameters may be vehicle speed data. When vehicle speed data is within a 5 mile per hour range for a steady-state determination time period of 10 seconds, the vehicle **110** may be determined to be in the steady-state condition. If the vehicle **110** is not determined to be at the steady-state condition, the algorithm **300** may return to or continue to determine the vehicle parameter at operation **310**.

[0025] If the vehicle **110** is determined to be at the steady-state condition, the algorithm **300** may proceed to measuring damping oscillation magnitude data at operation **320**. The damping oscillation magnitude data may be damping oscillation magnitude data collected from each respective one of the set of suspension dampers **160** within a damping oscillation magnitude data collection time period. The damping oscillation magnitude data collection time period may overlap with the steady-state determination time period partially, fully, or not at all. In some cases, the damping oscillation magnitude data collection time period may be a brief time period directly after the steady-state determination time period. The damping oscillation magnitude data collection time period may only include a single value, a plurality of values, or an average of a plurality of values of current suspension height data.

[0026] In an example embodiment, damping oscillation magnitude data may be determined based on the second data measured by the second sensor suite. In some cases, damping oscillation magnitude data may be directly determined based on the suspension height data. In an example embodiment, damping oscillation magnitude data may be determined based on suspension stroke position. Suspension stroke position may be defined by where specifically the suspension assembly is disposed at a given time and the direction the suspension assembly may be headed at that time. Suspension stroke position may be determined via rate of change of suspension velocity data and suspension height data.

[0027] After collection of the damping oscillation magnitude data, the algorithm **300** may perform a comparison between a reference suspension damper and remaining suspension dampers in operation **330** and may determine an operation state of the reference suspension damper at operation **340** based on the comparison performed in operation **330**. Later in the process, each one of the suspension dampers may take a turn being considered to be the reference suspension damper.

[0028] In an example embodiment, the comparison of the reference suspension damper to the remaining suspension dampers is comparing the damping oscillation magnitude data of the reference suspension damper to the damping oscillation magnitude data of the remaining suspension dampers. The reference suspension damper may be one of the set of suspension dampers **160** isolated for the remaining ones of the set of suspension dampers **160**. The remaining ones of the set of suspension dampers **160** may then be the remaining suspension dampers. In some cases, the damping oscillation magnitude data of the remaining suspension dampers may be an average of the damping oscillation magnitude data of all of the remaining suspension dampers. In an example embodiment, the damping oscillation magnitude data of the remaining suspension dampers may be a median of the damping oscillation magnitude data of all of the remaining suspension dampers. The damping oscillation magnitude data of the reference suspension damper may also be an average, a median, or any other grouping or mathematical process performed of the damping oscillation magnitude data of the reference suspension damper.

[0029] In some cases, the damping oscillation magnitude data of the remaining suspension dampers may be represented by a first tolerance range based on the damping oscillation magnitude data of all of the remaining suspension dampers. In an example embodiment, the first tolerance range may

be based on the average of the damping oscillation magnitude data of all of the remaining suspension dampers. In some cases, the first tolerance range may include a first tolerance range upper bound and a first tolerance range lower bound that are each separated from the average of the damping oscillation magnitude data of all of the remaining suspension dampers by a predetermined amount in opposing directions to define the first tolerance range. In an example embodiment, the first tolerance range upper bound may be a fixed percentage or fixed value above the average of the damping oscillation magnitude data of all of the remaining suspension dampers, and the first tolerance range lower bound may be a fixed percentage or fixed value below the average of the damping oscillation magnitude data of all of the remaining suspension dampers. For example, the first tolerance range upper bound may be 105% of the average of the damping oscillation magnitude data of all of the remaining suspension dampers and the first tolerance range lower bound may be 95% of the average of the damping oscillation magnitude data of all of the remaining suspension dampers. The first tolerance range may be related to the standard deviation of the damping oscillation magnitude data of the remaining suspension dampers. In some cases, the first tolerance range upper bound and the first tolerance range lower bound may be one standard deviation above and one standard deviation below, respectively, the average of the damping oscillation magnitude data of all of the remaining suspension dampers. In an example embodiment, the first tolerance range may be any range based on adjustments or functions applied damping oscillation magnitude data of all of the remaining suspension dampers that may determine an operation state of the reference suspension damper.

[0030] In some cases, the comparison may determine or classify the operation state as a selected one of a fault state **342** and a normal state **343**, as shown in FIG. **3**. In an example embodiment, the operation state may be determined to be a fault state **342** if the damping oscillation magnitude data of the reference suspension damper is outside of the first tolerance range. If the damping oscillation magnitude data of the reference suspension damper is within the first tolerance range, the operation state may be determined to be a normal state **343**.

[0031] The algorithm **300** may then determine if a final cycle has been completed at operation **350**. A final cycle may be determined to have been completed when every one of the set of suspension dampers **160** has had its operation state determined via being the reference suspension damper. In some cases, the algorithm **300** may cycle which of one the set of suspension dampers **160** is the reference suspension damper at operation **355**. In this regard, the reference suspension damper and remaining suspension dampers may be only temporary classifications of the individual suspension dampers of the set of suspension dampers **160**.

[0032] An example in accordance with an example embodiment may be seen in FIG. **4**. In this instance, the set of suspension dampers **160** may include a first suspension damper **161**, a second suspension damper **162**, a third suspension damper **163**, and a fourth suspension damper **164**. Initially, the algorithm **300** may determine that the first suspension damper **161** is the reference suspension damper and thus, the algorithm **300** may determine the second suspension damper **162**, the third suspension damper **163**, and the fourth suspension damper **164** as the remaining suspension dampers. In this regards, the operation state of the first suspension damper **161** may be determined by a comparison of the damping oscillation magnitude data of the first suspension damper **161** to the first tolerance range based on the damping oscillation magnitude data of the remaining suspension dampers (or the second suspension damper **162**, the third suspension damper **163**, and the fourth suspension damper **164**) in operations **330** and **340**. The temporary classification of the first suspension damper **161** as the reference suspension damper may be determined to be the first cycle **351**.

[0033] After the operation state of the reference suspension damper in the first cycle **351** has been determined, the algorithm **300** may determine that the final cycle has not been completed at operation **350** as the operation state of the second suspension damper **162**, the third suspension damper **163**, and the fourth suspension damper **164** has yet to be determined. Thus, the algorithm

300 may cycle the reference suspension damper at operation **355** to the second suspension damper **162**. As such, the remaining suspension dampers may be the first suspension damper **161**, the third suspension damper **163**, and the fourth suspension damper **164**. The temporary classification of the second suspension damper **162** as the reference suspension damper may be determined to be a second cycle **352**. The operation state of the second suspension damper **162** may be determined by a comparison of the damping oscillation magnitude data of the second suspension damper **162** to the first tolerance range based on the damping oscillation magnitude data of the remaining suspension dampers (or the first suspension damper **161**, the third suspension damper **163**, and the fourth suspension damper) in operations **330** and **340**.

[0034] The algorithm **300** may repeat operations **330**, **340**, **350**, and **355** until the operation state of every one of the set of suspension dampers **160** has been determined. In this regard, the operation state of the third suspension damper **163** may be determined in a third cycle **353**, and the operation state of the fourth suspension damper **164** may be determined in a fourth cycle **354**.

[0035] In an example embodiment, after the algorithm **300** may determine that the final cycle is completed in operation **350**, the algorithm **300** may check if the operation state of any one of the set of suspension dampers **160** is determined to be a fault state **342** in operation **360**. If none of the operation states of the set of suspension dampers **160** is determined to be a fault state **342**, the algorithm **300** may return to operation **310** and may determine or may continue to determine vehicle parameters. If the operation state of any one of the set of suspension dampers **160** is determined to be a fault state **342**, every instance of the fault state **342** may be classified in operation **370**.

[0036] In some cases, the fault state **342** of a suspension damper may be classified by the algorithm **300** according to the flow chart shown in FIG. 5. In this regard, the comparison of the reference suspension damper to the remaining suspension dampers performed in operation **330** may be revisited. The damping oscillation magnitude data of the reference suspension damper may not only be compared to the first tolerance range, but additionally to a preset deviation of the first tolerance range. In an example embodiment, the preset deviation may be a predetermined value or a variable value that may be applied to the first tolerance range upper bound and the first tolerance range lower bound. In some cases, the present deviation may be a fixed percentage, fixed value, or related to the standard deviation of the damping oscillation magnitude data of the remaining suspension dampers. In an example embodiment, the preset deviation may be a predetermined value set by the vehicle manufacturer, selected prior to any operation of the vehicle **110**, based on the vehicle classification and/or vehicle model.

[0037] In some cases, the fault state **342** may be classified as a selected one of a minor fault **371** or a major fault **372**. In an example embodiment, if the damping oscillation magnitude data of the reference suspension damper is outside first tolerance range but within the preset deviation of first tolerance range (as seen in operation **373**), the fault state may be classified as the minor fault **371**. In some cases, if the damping oscillation magnitude data of the reference suspension damper is outside the preset deviation of first tolerance range (as seen in operation **374**), the fault state may be classified as the major fault **372**.

[0038] In an example embodiment, algorithm **300** may report the fault state **342** in operation **380** upon determining and classifying the fault state **342** in operation **370**. Depending on the classification of the fault state **342**, the algorithm **300** may report the fault state **342** to different sources. In some cases, if the fault state **342** may be determined as a minor fault **371**, a report may be delivered to an operator of the vehicle **110** via a display or user interface **221** of the vehicle **110**. In an example embodiment, the report may be an auditory, visual, or haptic indication that may be transmitted by the communication module **220** of the controller **200** to a smart phone or other user device of the operator of the vehicle **110**. In some cases, the report may be transmitted by the communication module **220** to a communication or user device associated with a service technician **222**. In an example embodiment, the report may only be sent to the operator if the fault state **342**

may be classified as the minor fault **371**. In some cases, the report may be sent to both the operator and the service technician **222** if the fault state **342** may be classified as the major fault **372**. In an example embodiment, even if the fault state **342** may be determined as occurring, a report may be sent to both, one, or neither of the operator and service technician **222**. In some cases, the report sent to the service technician **222** may include vehicle specification and suspension damper specific details so that a replacement for the suspension damper or a specific component of the suspension damper may be pre-ordered. For example, the suspension damper specific details may be a suspension damper model or a suspension damper location.

[0039] In an example embodiment, the algorithm **300** may determine the operation state utilizing only onboard vehicle resources. For example, the operation state may be determined without the use of an external database or external processing. In some cases, an external database of previous suspension height data may be used by the controller **200**. In an example embodiment, the controller **200** may upload current suspension height data to an external controller. In an example embodiment, machine learning or iterative learning may be used by the controller **200** to optimize the algorithm **300**. In some cases, the first threshold range may be determined via machine learning.

[0040] In an example embodiment, the first tolerance range may only be one of a plurality of ranges. Each range of the plurality of ranges may be determined based on different periods of the damping oscillation magnitude data time period. For example, the plurality of ranges may include a daily range, a weekly range, a monthly range, and a lifetime range. The plurality of ranges may be based on any number of the damping oscillation magnitude data time periods. In some cases, the algorithm **300** may utilize the plurality of ranges to perform trend analysis on the set of suspension dampers **160**. Trend analysis may be performed by comparing the damping oscillation magnitude data of the reference suspension damper to the plurality of ranges, as well as comparing the plurality of ranges to one another. In an example embodiment, the operation state may be determined via trend analysis.

[0041] In some cases, the algorithm **300** may include a plurality of preset deviations. The plurality of preset deviations may be set by the vehicle's manufacture to define specific mechanical and electrical functioning limits of the vehicle **110**. The plurality of preset deviations may be used in trend analysis similar to the plurality of ranges.

[0042] In some cases, trend analysis may be performed using the plurality of ranges and the plurality of preset deviations together. In an example embodiment, the algorithm **300** may be run continuously and automatically by the controller **200**. In some cases, the any and all operations of the algorithm **300** may be run simultaneously and in real time. For example, operations **320**, **330**, **340**, **350**, **355**, **360**, and **370** may be run together to determine and classify the operation state of every one of the set of suspension dampers **160** at the same time. In this regards, the first cycle **351**, the second cycle **352**, the third cycle **353**, and the fourth cycle **354** may be run simultaneously in accordance with an example embodiment.

[0043] In an example embodiment, the result of the algorithm **300**, in the case of a fault state **342**, may be used compensate for suspension damper performance if the set of suspension dampers **160** are active suspension dampers by increasing the damping force of the individual suspension dampers to bring their performance in line with tolerances or performance of other suspension dampers on the vehicle. In some cases, this increase may be a temporary or may be more permanent solution depending on the severity of the fault state **342**. In an example embodiment, the increase may be useful for extending individual suspension damper life or providing a 'limp home' solution if the fault state **342** is classified as a major fault **372**.

[0044] FIG. **6** illustrates a block diagram of a method for determining an operation state of a set of suspension dampers in accordance with an example embodiment. The method may include receiving first data comprising driver input data and vehicle system data measured from a first sensor suite in operation **600**. The method may further include determining whether the vehicle is

in a steady state condition based on the first data in operation **610**. The method also may include, responsive to if in steady-state condition may be present, determining damping oscillation magnitude data for a set of suspension dampers based on second data comprising suspension assembly data measured from a second sensor suite in operation **620**. The method may include performing a comparison between the damping oscillation magnitude data for a reference suspension damper and the damping oscillation data for remaining suspension dampers in operation **630**. The method may continue to include determining an operation state of the reference suspension damper based on the comparison in operation **640**. The method may persist to include cycling the reference suspension damper to one of the remaining suspension dampers in operation **650**. Finally, the method may include repeating performing the comparison, determining the operation state of the reference suspension damper, and cycling of the reference suspension damper until the operation state of each one of a set of suspension dampers has been determined. at operation **660**.

[0045] A control system for a suspension system of a vehicle may be provided. The control system may include a set of suspension dampers that may be operably coupled to a set of wheel assemblies of the vehicle, a first sensor suite that may measure first data that may include driver input data and vehicle system data, a second sensor suite that may measure second data that may include suspension assembly data, and a controller that may store the first data that may be received from the first sensor suite and the second data that may be received from the second sensor suite. The set of suspension dampers may include a reference suspension damper that may be operably coupled to a corresponding one of the set of wheel assemblies and remaining suspension dampers that may be operably coupled to remaining corresponding ones of the set of wheels. The controller may be configured to receive the first data, determine whether the vehicle may be in a steady state condition based on the first data, and responsive to the vehicle being determined to be in the steady state condition, determine damping oscillation magnitude data for the set of suspension dampers based on the second data. The controller may be further configured to perform a comparison between the damping oscillation magnitude data for the reference suspension damper and the damping oscillation data for the remaining suspension dampers, determine an operation state of the reference suspension damper based on the comparison, cycle the reference suspension damper to the one of the remaining suspension dampers; and repeat performing the comparison, determining the operation state of the reference suspension damper, and cycling of the reference suspension damper until the operation state of each one of the set of suspension dampers has been determined.

[0046] The control system for a suspension system of a vehicle of some embodiments may include additional features, modifications, augmentations and/or the like to achieve further objectives or enhance performance of the suspension assembly. The additional features, modifications, augmentations and/or the like may be added in any combination with each other. Below is a list of various additional features, modifications, and augmentations that can each be added individually or in any combination with each other. For example, the comparison may classify the operation state as a selected one of a normal state and a fault state. In an example embodiment, responsive to the damping oscillation magnitude data of the reference suspension damper being within a first tolerance range of the damping oscillation magnitude data of the remaining suspension dampers, the operation state may be determined to be a normal state. In some cases, the first tolerance range may be based on an average of the damping oscillation magnitude data of the remaining suspension dampers. In an example embodiment, responsive to the damping oscillation magnitude data of the reference suspension damper being outside the first tolerance range of the damping oscillation magnitude data of the remaining suspension dampers, the operation data may be determined to be fault state. In some cases, responsive to the operation data being determined to be fault state, the controller may be configured to classify the fault state as a minor fault or a major fault based on the comparison. In an example embodiment, responsive to the damping oscillation magnitude data of the reference suspension damper being outside the first tolerance range of the damping oscillation

magnitude data of the remaining suspension dampers but within a preset deviation from the first tolerance range, the fault state may be classified as the minor fault. In some cases, responsive to the damping oscillation magnitude data of the reference suspension damper being outside the preset deviation from the first tolerance range, the fault state may be classified as the major fault. In an example embodiment, the preset deviation may be based on vehicle classification or vehicle model prior to any operation of the vehicle. In some cases, responsive to determining the operation state is the fault state, the controller is configured to report details of the fault state to an operator of the vehicle via a user interface or to a service technician based on how the fault state is classified. In an example embodiment, wherein responsive to fault state being classified as the major fault, the controller may be configured to report to a service technician the details of the fault state and suspension damper specific details, and the suspension damper specific details may include suspension damper model and suspension damper location. In some cases, the suspension assembly data may be a voltage measurement from a suspension height sensor. In an example embodiment, the driver input data may include throttle inputs, brake inputs, and steering inputs, and the vehicle system data may include inertia sensor data and tire pressure.

[0047] A method for determining an operation state of a suspension damper of a vehicle may therefore be provided. The method may include receiving first data that may include driver input data and vehicle system data that may be measured from a first sensor suite, determining whether the vehicle may be in a steady state condition based on the first data, and responsive to the vehicle being determined to be in the steady state condition, determining damping oscillation magnitude data for a set of suspension dampers based on second data that may include suspension assembly data that may be measured from a second sensor suite. The method may further include performing a comparison between the damping oscillation magnitude data for a reference suspension damper and the damping oscillation data for remaining suspension dampers, determining an operation state of the reference suspension damper based on the comparison, cycling the reference suspension damper to one of the remaining suspension dampers, and repeating performing the comparison, determining the operation state of the reference suspension damper, and cycling of the reference suspension damper until the operation state of each one of a set of suspension dampers has been determined. The set of suspension dampers may be operably coupled to a set of wheel assemblies of the vehicle. The set of suspension dampers may include the reference suspension damper that may be operably coupled to a corresponding one of the set of wheel assemblies and the remaining suspension dampers that may be operably coupled to remaining corresponding ones of the set of wheels.

[0048] Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe exemplary embodiments in the context of certain exemplary combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. In cases where advantages, benefits or solutions to problems are described herein, it should be appreciated that such advantages, benefits and/or solutions may be applicable to some example embodiments, but not necessarily all example embodiments. Thus, any advantages, benefits or solutions described herein should not be thought of as being critical, required or essential to all embodiments or to that which is claimed herein. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

- 1.** A control system for a suspension system of a vehicle, the control system comprising: a set of suspension dampers operably coupled to a set of wheel assemblies of the vehicle; a first sensor suite for measuring first data comprising driver input data and vehicle system data; a second sensor suite for measuring second data comprising suspension assembly data; a controller storing the first data received from the first sensor suite and the second data received from the second sensor suite, wherein the set of suspension dampers comprises a reference suspension damper operably coupled to a corresponding one of the set of wheel assemblies and remaining suspension dampers operably coupled to remaining corresponding ones of the set of wheel assemblies; wherein the controller is configured to: receive the first data; determine whether the vehicle is in a steady state condition based on the first data; responsive to the vehicle being determined to be in the steady state condition, determine damping oscillation magnitude data for the set of suspension dampers based on the second data, perform a comparison between the damping oscillation magnitude data for the reference suspension damper and the damping oscillation data for the remaining suspension dampers, determine an operation state of the reference suspension damper based on the comparison, cycle the reference suspension damper to the one of the remaining suspension dampers; repeat performing the comparison, determining the operation state of the reference suspension damper, and cycling of the reference suspension damper until the operation state of each one of the set of suspension dampers has been determined.
- 2.** The control system of claim 1, wherein the comparison classifies the operation state as a selected one of a normal state and a fault state.
- 3.** The control system of claim 2, wherein responsive to the damping oscillation magnitude data of the reference suspension damper being within a first tolerance range of the damping oscillation magnitude data of the remaining suspension dampers, the operation state is determined to be a normal state.
- 4.** The control system of claim 3, wherein the first tolerance range is based on an average of the damping oscillation magnitude data of the remaining suspension dampers.
- 5.** The control system of claim 3, wherein responsive to the damping oscillation magnitude data of the reference suspension damper being outside the first tolerance range of the damping oscillation magnitude data of the remaining suspension dampers, the operation state is determined to be fault state.
- 6.** The control system of claim 5, wherein responsive to the operation state being determined to be fault state, the controller is configured to classify the fault state as a minor fault or a major fault based on the comparison.
- 7.** The control system of claim 6, wherein responsive to the damping oscillation magnitude data of the reference suspension damper being outside the first tolerance range of the damping oscillation magnitude data of the remaining suspension dampers but within a preset deviation from the first tolerance range, the fault state is classified as the minor fault.
- 8.** The control system of claim 7, wherein responsive to the damping oscillation magnitude data of the reference suspension damper being outside the preset deviation from the first tolerance range, the fault state is classified as the major fault.
- 9.** The control system of claim 7, wherein the preset deviation is based on vehicle classification or vehicle model prior to any operation of the vehicle.
- 10.** The control system of claim 6, wherein responsive to determining the operation state is the fault state, the controller is configured to report details of the fault state to an operator of the vehicle via a user interface or to a service technician based on how the fault state is classified.
- 11.** The control system of claim 10, wherein responsive to fault state being classified as the major fault, the controller is configured to report to a service technician the details of the fault state and

suspension damper specific details, wherein the suspension damper specific details comprise suspension damper model and suspension damper location.

12. The control system of claim 1, wherein the suspension assembly data is a voltage measurement from a suspension height sensor.

13. The control system of claim 1, wherein the driver input data further comprises throttle inputs, brake inputs, and steering inputs, and wherein the vehicle system data further comprises inertia sensor data and tire pressure.

14. A method for determining an operation state in a suspension damper of a vehicle, the method comprising: receiving first data comprising driver input data and vehicle system data measured from a first sensor suite; determining whether the vehicle is in a steady state condition based on the first data; responsive to the vehicle being determined to be in the steady state condition, determining damping oscillation magnitude data for a set of suspension dampers based on second data comprising suspension assembly data measured from a second sensor suite, performing a comparison between the damping oscillation magnitude data for a reference suspension damper and the damping oscillation data for remaining suspension dampers, determining an operation state of the reference suspension damper based on the comparison, cycling the reference suspension damper to one of the remaining suspension dampers; repeating performing the comparison, determining the operation state of the reference suspension damper, and cycling of the reference suspension damper until the operation state of each one of a set of suspension dampers has been determined, wherein the set of suspension dampers is operably coupled to a set of wheel assemblies of the vehicle, and wherein the set of suspension dampers comprises the reference suspension damper operably coupled to a corresponding one of the set of wheel assemblies and the remaining suspension dampers operably coupled to remaining corresponding ones of the set of wheel assemblies.

15. The method of claim 14, wherein the comparison classifies the operation state as a selected one of a normal state and a fault state.

16. The method of claim 15, wherein responsive to the damping oscillation magnitude data of the reference suspension damper being within a first tolerance range of the damping oscillation magnitude data of the remaining suspension dampers, the operation state is determined to be a normal state.

17. The method of claim 16, wherein responsive to the damping oscillation magnitude data of the reference suspension damper being outside the first tolerance range of the damping oscillation magnitude data of the remaining suspension dampers, the operation state is determined to be fault state.

18. The method of claim 17, wherein responsive to the operation state being determined to be fault state, the fault state is classified as a minor fault or a major fault based on the comparison.

19. The method of claim 18, wherein responsive to the damping oscillation magnitude data of the reference suspension damper being outside the first tolerance range of the damping oscillation magnitude data of the remaining suspension dampers but within a preset deviation from the first tolerance range, the fault state is classified as the minor fault.

20. The method of claim 19, wherein responsive to the damping oscillation magnitude data of the reference suspension damper being outside the preset deviation from the first tolerance range, the fault state is classified as the major fault.
